

Benchmarking LLM-driven Model evasion attacks against NIDS

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30th April 2026

1 Working title

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2 Background

Traditional Network Intrusion Detection Systems (NIDS), primarily relying on signature-based and anomaly-based systems, remain a cornerstone of digital infrastructure defense [1]. However, the emergence of Large Language Models (LLMs) introduces a sophisticated vector for automating Model Evasion Attacks. These models can be leveraged to generate polymorphic malicious traffic that mimics legitimate patterns, specifically designed to bypass the static detection logic of conventional NIDS.

Currently, there is a lack of systematic, and widely adopted frameworks dedicated to evaluating the effectiveness of adversarial traffic generated by LLMs on various established NIDS architectures. This research focuses on developing a benchmark to quantify this threat, providing a baseline for understanding how generative AI can compromise the security layers of legacy networks.

3 Research question

To what extent can Large Language Models be used to modify malicious traffic in order to reduce the detection rate of attacks by Network Intrusion Detection Systems, compared to the number of attacks detected by the same type of systems when pre-modified traffic is used to carry out the same of attacks?

4 Hypothesis

The hypothesis that this master's thesis aims to validate is that by using LLMs, malicious attack traffic can be modified in such a way that the evasion/detection rate of Network Intrusion Detection Systems changes significantly.

5 Research method

The research methodology will be structured into these four main phases:

1. **Literature Review:** Systematic collection of existing and updated NIDS evasion techniques, which includes identifying common signatures in traditional NIDS that are susceptible to polymorphic transformations, as well as current state-of-the-art methods for generating synthetic network traffic using Large Language Models (LLMs).
2. **Benchmark Design:** Defining the experimental environment, which includes selecting diverse NIDS implementations (e.g., Snort and Suricata), choosing appropriate legit and malicious traffic datasets, and identifying the LLM architectures to be used for generating adversarial model evasion attacks.
3. **Experimental Execution:** Implementing the attack generation pipeline where LLMs are prompted to generate different malicious payloads. These variants will be played against the selected NIDS implementations to measure their bypass success rates under controlled conditions.
4. **Comparative Analysis:** Statistical evaluation of the results to quantify the decrease in detection accuracy on each one of the NIDS implementations. This phase will involve comparing the effectiveness of different LLM models (varying on training and architecture) on the payload generation.

A more detailed and technical explanation of sections 2, 3 and 4 is available in the *Individual Plan* document.

6 Background of the student

My academic background includes core courses that provide the theoretical and practical basis for this project: **Language-Based Security (DD2459)**, which is essential for understanding injection vulnerabilities and information flow; **Ethical Hacking (DA2210)**, which provided a hands-on perspective on exploit development; and **Building Networked Systems Security (EP2510)**.

My Master's thesis is being conducted at RISE (Research Institutes of Sweden), under the supervision of Han Wang, PhD in Computer Science and researcher at this institution. This collaboration provides me with access to specialized expertise in the intersection of Machine Learning and Network Security. Working within the RISE ecosystem allows me to leverage state-of-the-art research resources and professional guidance.

7 Supervisor at the company/external organization

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8 Suggested examiner at KTH

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9 Suggested supervisor at KTH

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10 Resources

The project will be conducted in collaboration with RISE. The development and evaluation of the benchmark will utilize the following resources:

- **Software:** Implementation of traditional NIDS environments using open-source tools such as *Snort* and *Suricata*.
- **Data Resources:** Access to standardized network traffic datasets containing both legitimate and malicious traffic for baseline evaluations.
- **LLM Infrastructure:** Use of state-of-the-art Large Language Models and the hardware acceleration required for generating adversarial payloads. RISE will provide access to the necessary computational infrastructure for LLM inference and fine-tuning.

The following literature will serve as the primary source of information for establishing the theoretical framework (see References section for full citations).

- Adversarial model evasion attacks in the networking domain focused on classical NIDS [2].
- Reliable datasets for training in the cybersecurity domain [3]
- LLM-based synthetic traffic generation and polymorphic payload transformation [4].
- Methodologies for evaluating NIDS resilience against evasion attacks [1].

11 Eligibility

At the time of writing this proposal, I have completed the following courses:

- **EN2720** Ethical Hacking (7.5 hp)
- **DD2391** Cybersecurity Overview (7.5 hp)
- **DD2510** Cybersecurity in a Socio-Technical Context (7.5 hp)
- **EP2500** Networked Systems Security (7.5 hp)
- **DD2520** Applied Cryptography (7.5 hp)
- **EP2520** Building Networked Systems Security (7.5 hp)
- **ID2218** Design of Fault-tolerant Systems (7.5 hp)
- **DA2215** Theory of Science and Scientific Methods in Cybersecurity (3.0 hp)
- **DD2525** Language-Based Security (7.5 hp)
- **AK2030** Theory and Methodology of Science (Natural and Technological Science) (4.5 hp)

- **EP2780** Digital Forensics and Incident Response (7.5 hp)

The total number of credits approved amounts to 77.5 ECTS. Also, at the same time, I am waiting to find out my grades, or I am taking the following subjects in P3 or P4:

- **LS1502** Swedish A1 for Engineers (7.5 hp)
- **DH2642** Interaction Programming and the Dynamic Web (7.5 hp)
- **DD2303** The Cybersecurity Engineer's Role in Society (0.5 hp remaining)

12 Study Planning

I have already completed 77.5 ECTS credits of the program. For the upcoming periods (P3 and P4), my only remaining course is **DD2303 The Cybersecurity Engineer's Role in Society (2.0 hp)**, which is a mandatory course on each period of the master's program.

References

- [1] P. Shetty, A. Meshram, M. Karch, C. Haas, and N. O. Tippenhauer, "Faser-in: Evasion of network intrusion detection systems in industrial networks," in *ACM CPSS 2025 - Proceedings of the 11th ACM Cyber-Physical System Security Workshop*. New York, NY, USA: ACM, 2025. ISBN 9798400714139 pp. 87–102. 1, 3
- [2] H. Kılıç, N. S. Katal, and A. A. Selçuk, "Evasion techniques efficiency over the ips/ids technology," in *2019 4th International Conference on Computer Science and Engineering (UBMK)*, 2019. doi: 10.1109/UBMK.2019.8907177 pp. 542–547. 3
- [3] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, "Edge-iiotset: A new comprehensive realistic cyber security dataset of iot and iiot applications: Centralized and federated learning," 2022. [Online]. Available: <https://dx.doi.org/10.21227/mbc1-1h68> 3
- [4] P. Sun, X. Yun, S. Li, T. Yin, C. Si, and J. Xie, "Advtg: An adversarial traffic generation framework to deceive dl-based malicious traffic detection models," *Proceedings of the ACM on Web Conference 2025*, 2025. [Online]. Available: <https://api.semanticscholar.org/CorpusID:277998670> 3

The approximate word count is 943 .

Acronyms

- **NIDS:** Network Intrusion Detection System
- **LLM:** Large Language Model
- **MEA:** Model Evasion Attacks